

Jeremy McMahan

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Research Interests

Topics *Algorithms under Uncertainty; Approximation Algorithms; Online, Stochastic and Combinatorial Optimization; Algorithmic Game Theory; Reinforcement Learning; Learning-Augmented Algorithms*

Summary Studying the **tractability of sequential decision-making** in *combinatorial* and *strategic* settings under *uncertainty*. My work designs polynomial-time approximation and learning algorithms with formal guarantees for structured sequential decision problems, spanning constrained control, multi-agent systems, and online optimization.

Education

- 2018–2025 **PhD in Computer Science**, *University of Wisconsin-Madison*
- Thesis: Safe Multi-Agent Reinforcement Learning in Polynomial Time
 - Advisor: Prof. Jerry Zhu
 - LEAP Fellow
- 2018–2020 **MS in Computer Science**, *University of Wisconsin-Madison*
- Thesis: A D-competitive algorithm for the Multilevel Aggregation Problem with Deadlines
 - Advisor: Prof. Shuchi Chawla
- 2016–2018 **BS in Mathematics and Computer Science**, *University of Illinois Urbana-Champaign*
- Thesis: Spectral Graph Isomorphism
 - Advisor: Alexandra Kolla
 - **Student ranked #1 in Mathematics and Computer Science**

Research Experience

- 2020–2025 **Graduate Research Assistant**, *University of Wisconsin-Madison*
- 2022–2024 **IFDS Research Assistantship**, *Institute for Foundations of Data Science*
Interdisciplinary research on the theoretical foundations of Data Science.

Core Research Contributions

- **Constrained Markov Decision Processes:** Resolved the long-standing polynomial-time approximability of general constrained MDPs by developing their first polynomial-time approximation schemes, enabling tractable decision-making under safety and resource constraints. (*ICML '25, NeurIPS '24, AISTATS '24*)

- **Robust Markov Games:** Resolved the tractability of equilibria robust to model uncertainty and environmental manipulation by developing their first polynomial-time algorithms, enabling reliable and incentive-aware multi-agent decision-making. (*ICML '25, ICML '24, AAAI '24*)
- **Behavior Influence in Markov Games:** Established the first polynomial-time algorithms for computing optimal online manipulation, offline reward poisoning, and planning misinformation attacks, enabling adversarial robustness evaluation and behavior influence in multi-agent decision systems. (*ICML '24, RLC '24, AAAI '24, AAAI '23*)
- **Online and Adaptive Optimization:** Established the state-of-the-art competitive and adaptive approximation guarantees for multi-level aggregation and correlated Pandora's Box, advancing tractable optimization for large-scale decision-making under uncertainty. (*APPROX '23, Preprint*)

Publications

**Indicates equal contribution. Authorship order follows venue conventions.*

Selected Publications

- ICML '25 **Polynomial-Time Approximability of Constrained Reinforcement Learning**
Jeremy McMahan
- First polynomial-time $(0, \epsilon)$ -bicriteria approximation for general constrained MDPs, the best approximation guarantee possible assuming $P \neq NP$.
 - Establishes a unifying framework for tractable constrained decision-making, including almost-sure, chance, and expectation constraints.
- NeurIPS '24 **Deterministic Policies for Constrained Reinforcement Learning in Polynomial Time**
Jeremy McMahan
- First FPTAS computing deterministic policies in single-constraint MDPs.
 - Resolved the long-standing existence of efficient approximations question for almost-sure, anytime, and deterministic-policy expectation constraints.
- ICML '24 **Roping in Uncertainty: Robustness and Regularization in Markov Games**
Jeremy McMahan, Giovanni Artiglio, Qiaomin Xie
- First characterizations and polynomial-time algorithms for equilibria robust to model-uncertainty in Markov games.
- AAAI '24 **Optimal Attack and Defense for Reinforcement Learning**
Jeremy McMahan, Young Wu, Xiaojin Zhu, Qiaomin Xie
- First characterizations and polynomial-time algorithms for optimal manipulation attacks and robust equilibrium defenses in Markov games.
 - Yields tractable methods for evaluating and ensuring policy robustness to adversarial environmental manipulation.
- APPROX '23 **Approximating Pandora's Box with Correlations**
Shuchi Chawla, Evangelia Gergatsouli*, Jeremy McMahan*, Christos Tzamos**
- First polynomial-time adaptive approximation for Pandora's Box with correlated values; remains the best-known approximation ratio.

- Preprint **A D-competitive Algorithm for the Multilevel Aggregation Problem with Deadlines**
Jeremy McMahan
○ First D -competitive algorithm for multi-level aggregation with deadlines; remains the best-known competitive ratio.
- Additional Refereed Publications**
- ICML '25 **Anytime-Constrained Equilibria in Polynomial Time**
Jeremy McMahan
- RLC '24 **Inception: Efficiently Computable Misinformation Attacks on Markov Games**
Jeremy McMahan, Young Wu, Yudong Chen, Xiaojin Zhu, Qiaomin Xie
- ICML '24 **Minimally Modifying a Markov Game to Achieve any Nash Equilibrium and Value**
Young Wu, Jeremy McMahan, Yiding Chen, Yudong Chen, Xiaojin Zhu, Qiaomin Xie
- CogSci '24 **Various Misleading Visual Features in Misleading Graphs: Do they truly deceive us?**
Jihyun Rho, Martina A Rau, Shubham Kumar Bharti, Rosanne Luu, Jeremy McMahan, Andrew Wang, Jerry Zhu
- AISTATS '24 **Anytime-Constrained Reinforcement Learning**
Jeremy McMahan, Xiaojin Zhu
- AAAI '24 **Data Poisoning to Fake a Nash Equilibrium for Markov Games**
Young Wu, Jeremy McMahan, Xiaojin Zhu, Qiaomin Xie
- AAAI '23 **Reward Poisoning Attacks on Offline Multi-Agent Reinforcement Learning**
Young Wu, Jeremy McMahan, Xiaojin Zhu, Qiaomin Xie
- GAMES '20 **Noble Deceit: Optimizing Social Welfare for Myopic Multi-Armed Bandits**
Ashwin Maran, Jeremy McMahan*, Nathaniel Sauerberg**

Research Software

- Python **Anytime-Constrained Decision-Making**, [[GitHub Link](#)]
An algorithmic framework for constrained decision-making, implementing polynomial-time optimization solvers and deep reinforcement learning approaches for anytime-constrained MDPs. Includes a comprehensive benchmarking suite for evaluating performance tradeoffs. (AISTATS '24)
- Python **Robust Multi-Agent Decision-Making**, [[GitHub Link](#)]
Reference implementations of polynomial-time algorithms for online attacks and defenses in Markov games, enabling empirical evaluation and equilibrium computation for robust multi-agent decision systems. (AAAI '24)
- Python **Knapsack Approximations Suite**, [[GitHub Link](#)]
A benchmarking and evaluation framework for approximation algorithms, comparing classical polynomial-time methods against deep-learning-based heuristics on large-scale resource allocation problems.

Selected Talks

From Knapsacks to Self Driving: FPTAS Recipes for Constrained Reinforcement Learning

- UMD College Park (MARL Seminar) *March 2025*
- UW-Madison (Theory of Computing Seminar) *November 2024*

Optimal Attack and Defense for Reinforcement Learning

- UW-Madison (Theory of Computing Seminar) *February 2025*
- UW-Madison (Computer Vision Roundtable) *January 2024*

Teaching Experience

Primary Instructor

- Summer 2022 **CS 577: Introduction to Algorithms**, *University of Wisconsin-Madison*
- Responsible for curriculum design, lecture delivery, course structure, and student evaluation for a class of 100 students.
 - Designed original lecture slides, lecture notes, lecture videos, and problem sets focusing on the design of efficient algorithms and rigorous proofs of correctness.
 - Managed one TA and grader.

Head Teaching Assistant

- Spring 2021, 2022 **CS 540: Introduction to Artificial Intelligence**, *University of Wisconsin-Madison*
- Coordinated the administrative and grading workflow for a large-scale course with 500 students, 8 TAs, and over 20 graders and peer mentors.
 - Developed standardized grading rubrics and course structures in addition to the standard teaching assistant duties.
 - Instructors: Ilias Diakonikolas, Sharon Li, Fred Sala, and Jerry Zhu

Guest Lecturer

- Spring 2025 **CS 839: Game Theory, Optimization & Learning**, *University of Wisconsin-Madison*
- Designed and presented two lectures on the theory of multi-agent reinforcement learning.
 - Instructor: Manolis Vlatakis
- Summer 2021, 2022 **New Horizons in Theoretical Computer Science Summer School**, *ACM SIGACT*
- Designed lectures spanning complexity theory basics, interactive proofs, and Markov-chain Monte Carlo methods.
 - Instructors: Antonio Blanca, Nicole Immorlica, Yael Kalai, Jelani Nelson, and Ashia Wilson

Graduate Teaching Assistant

- Fall 2020, 2021 **CS 540: Introduction to Artificial Intelligence**, *University of Wisconsin-Madison*
- Instructors: Anthony Gitter, Josiah Hanna, Yin Li, Yingyu Liang, and Daifeng Wang

- Spring 2019, **CS 577: Introduction to Algorithms**, *University of Wisconsin-Madison*
Instructors: Shuchi Chawla, Dieter van Melkebeek, and Christos Tzamos
- Fall 2019 **CS 301: Introduction to Data Science**, *University of Wisconsin-Madison*
Instructor: Tyler Caraza-Harter
- Fall 2018 **CS 524: Introduction to Optimization**, *University of Wisconsin-Madison*
Instructor: Steve Wright

Awards and Honors

- 2021–2025 **LEAP Fellow**, *CMD-IT / NSF Alliance*
Selected for the NSF-funded Leadership, Excellence, Access, and Pathways (LEAP) Alliance fellowship, focused on diversifying future faculty leadership in computing.
- 2023 **Graduate Instructor Award**, *University of Wisconsin-Madison*
Recognition for excellent performance as a primary instructor.
- 2019–2022 **The "Golden Brick" Teaching Award**, *University of Wisconsin-Madison*
A student/instructor-voted honor for most effective teaching. Won consecutively for every year served as a TA (4 years).
- 2018–2019 **CS Departmental Scholarship**, *University of Wisconsin-Madison*
Merit-based scholarship awarded to incoming PhD students.
- 2018 **Most Outstanding Major Award**, *University of Illinois Urbana-Champaign*
Awarded to the #1-ranked student in Mathematics and Computer Science.

Professional Service

- 2024–2025 **Conference Reviewer**
NeurIPS (2024, 2025), ICML (2024, 2025), AISTATS (2025), RLC (2024)

Leadership and Organization

- 2023–2025 **Co-Organizer**, *Machine Learning Lunch Meetings (MLLM) Seminar, University of Wisconsin-Madison*
Organized weekly research seminars for the ML community, which included managing speaker invitations, scheduling, and logistics for internal and external presenters.

Mentorship and Outreach

- 2021–2022 **Planning Committee & Teaching Assistant**, *ACM SIGACT, New Horizons in Theoretical Computer Science Summer School*
A week-long summer school aimed at inspiring undergraduates from under-represented backgrounds to pursue careers in TCS.
- Served on the planning committee.
 - Mentored students during technical sessions and social events.
 - Developed and presented lessons on selected TCS topics.

Technical Skills

- Languages **Python, Julia, C/C++, Mathematica, SageMath, L^AT_EX, Bash**
Libraries **Gymnasium, Gurobi, PyTorch, SciPy, NumPy, Pandas**